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Internship

AI Intern **Feb 2026 - Present**
Infosys Springboard

- Analyzed **multi-dimensional sales datasets using SQL and Pandas** to optimize strategies for small businesses, identifying **3+ key profitability drivers and operational gaps**.
- Built a **robust data pipeline** involving data cleaning, preprocessing, and Exploratory Data Analysis (EDA) using **NumPy and Pandas**, improving dataset quality and reliability for modeling.
- Developed **interactive visualizations and profit analysis reports** using **Matplotlib and Seaborn**, transforming complex datasets into actionable insights for decision-making.
- Implemented **machine learning forecasting models** to predict future sales trends, improving forecasting accuracy and helping businesses plan upcoming quarters effectively.

Projects

AI-Powered Exam Proctoring / *OpenCV, NumPy, Pandas, SQL, Machine Learning* **Feb 2022 - March 2025**

- Built a **Computer Vision pipeline using OpenCV** for real-time anomaly detection and behavioral feature extraction during online exams.
- Developed a **probabilistic risk-scoring engine** combining signals such as **head movement and eye tracking** to detect suspicious behavior while minimizing false positives.
- Designed a real-time monitoring dashboard using HTML/CSS to display risk indicators and automated alerts for proctors.
- Validated model performance through **cross-validation techniques**, ensuring reliability across diverse environments.
- Github Repository Link:

Detect traffic signs with deep learning / *TensorFlow, Keras, OpenCV, Pandas* **Jan 2026 - Feb 2026**

- Built a **Convolutional Neural Network (CNN)** using **Keras and TensorFlow** to classify traffic signs from images.
- Preprocessed image datasets using **OpenCV and NumPy**, including resizing, normalization, and augmentation to improve model performance.
- Trained and evaluated the model on a traffic sign dataset achieving high classification accuracy.
- Implemented **real-time traffic sign detection** by integrating the trained model with a webcam/video feed.
- Github Repository Link:

Certificates

Cloud Computing **October 2025**
NPTEL — *Link*

Advanced Python for AI and ML (Summer Training) **June 2025**
CSE Pathshala — *Link*

Technical Skills

Languages: C++, Java, Python, SQL, Bash/Shell Scripting
Technologies/Frameworks: **Scikit-learn**, **TensorFlow**, PyTorch, **Keras**, OpenCV, **XGBoost**
Tools & Platforms: Git, GitHub, **MySQL**, **Linux (Ubuntu)**, **Jupyter Notebook**, **Power BI**, **AWS**, **Hadoop**, **Spark**
MLOps & Deployment: Docker, **FastAPI**, **Flask**, **Streamlit**, Model Deployment, REST APIs
Skills: Data Structures and Algorithms, Problem-Solving, Responsive Web Design, Scripting in Python and JavaScript

Achievement

Innovate X Hackathon – Top 10 Finalist:
Reached the final round in the Innovate X Hackathon out of 120 teams.

Data Structures & Algorithms:
Solved **300+ algorithmic problems** on LeetCode and GeeksforGeeks covering arrays, strings, linked lists, stacks, queues, trees, graphs, dynamic programming and strong grasp of time–space complexity.

Competitive Programming: Global Rank 252 in LeetCode Weekly Contest 468 (34k+ participants).

Education

Lovely Professional University **2023 – Present**
Computer Science and Engineering — CGPA: 8.25 *Jalandhar, Punjab*

Manisha International School **2022 – 2023**
12th with Science — Percentage: 82.60% *Durgapur, West Bengal*

Satyanarayan Academy **2020 – 2021**
10th with Science — Percentage: 80% *Durgapur, West Bengal*